University of Cape Town

Department of Statistical Sciences

Introduction to Bayes Assignment - 2026

Instructions:

- This assignment is to be completed in groups of up to three members.
- The assignment will contain some concepts that we have not formally covered in class. You will be expected to do some research on these aspects of the assignment. Give yourself sufficient time to complete the assignment. Do not start a day or two before the hand in date.
- Late hand-ins will be penalised by 10% per day late.
- An electronic copy of the assignment as well as your (working) R code should be submitted by 11.55pm on the 20th of April.
- This electronic copy must be a LaTeX-based document. In other words, the write up should be in Markdown, Quarto, or LaTeX. Any handwritten components or Microsoft Word documents converted to pdf will be penalised.
- There may be bonus marks available, but the maximum you may earn is 100%.
- Include your student number on the hand-in and remember to sign a plagiarism declaration.

Preamble

We will use the **Auto MPG** dataset, which contains fuel consumption data for various vehicles. The goal is to model miles per gallon (`mpg`) as a function of vehicle characteristics. Use the code below to load and prepare the data. Note that scaling has been applied on the predictors.

```
# Packages required
require(MASS)

# Load the Auto MPG dataset
auto <- read.csv(
   https://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/auto-mpg.data ,
  header = FALSE, sep =   ,
  col.names = c( mpg , cylinders , displacement , horsepower ,
                 weight , acceleration , model_year , origin , car_name ),
  na.strings =  ?
)
auto <- na.omit(auto)

Y <- matrix(auto$mpg, ncol = 1)
Xpreds <- auto[, c( displacement ,  horsepower ,  weight ,  acceleration )]
X <- as.matrix(cbind(1, scale(Xpreds)))
n <- nrow(X)
```

Q1) Linear Regression with Gibbs Sampling [30 marks]

Consider the linear regression model, $y_i = \mathbf{x}_i^T \boldsymbol{\beta} + e_i$ where $e_i \sim N(0, \sigma^2)$ for $i = 1, \ldots, n$.

Assume that we have the following priors:

$$\boldsymbol{\beta} \mid \sigma^2 \sim N_{k+1}\left(\tilde{\boldsymbol{\beta}},\ \sigma^2 \mathbf{M}\right) \quad \text{and} \quad \sigma^2 \sim \mathrm{IG}(a, b), \tag{A1}$$

where $\tilde{\boldsymbol{\beta}} = \mathbf{0}$, $\mathbf{M} = \mathbf{I}_{k+1}$ and $a = b = 1$.

(a) Using the assumptions from eq. (A1), **clearly** show that the conditional posterior distributions are given as:

find $\beta|\sigma^2$ & $\sigma^2|y$

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}, \mathbf{X} \sim N_{k+1}\left(\boldsymbol{\mu}_\beta,\ \sigma^2 \left(\mathbf{M} + \mathbf{X}^T\mathbf{X}\right)^{-1}\right)$$

$$\sigma^2 \mid \mathbf{y}, \mathbf{X} \sim \mathrm{IG}\left(a + \frac{n}{2},\ b + \frac{A_2}{2}\right)$$

where

$$\boldsymbol{\mu}_\beta = \left(\mathbf{M} + \mathbf{X}^T\mathbf{X}\right)^{-1}\left(\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} + \mathbf{M}\tilde{\boldsymbol{\beta}}\right)$$

$$A_2 = \mathbf{y}^T\mathbf{y} + \tilde{\boldsymbol{\beta}}^T\mathbf{M}\tilde{\boldsymbol{\beta}} - \boldsymbol{\mu}_\beta^T\left(\mathbf{M} + \mathbf{X}^T\mathbf{X}\right)\boldsymbol{\mu}_\beta.$$

Note that $\hat{\boldsymbol{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y}$.

Hint:

- $[\sigma^2 \mid \mathbf{y}, \mathbf{X}] = \int_{\boldsymbol{\beta}} [\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, \mathbf{X}] d\boldsymbol{\beta}$ integrate β out.
- $|a\mathbf{A}| = a^k|\mathbf{A}|$, where $\mathbf{A}$ is a k by k matrix.

① Find $\beta, \sigma^2 | y, X$

② Integrate β out.

[5+5 marks]

(b) Using the results from above, write R code to generate posterior samples. [5 marks]

(c) Using the function above, obtain at least 50 000 posterior samples.

(i) Create trace plots for each of the regression coefficients. Briefly discuss the results and what they mean. [2+1 marks]

(ii) Create density plots for each of the regression coefficients.

i. Show where the posterior mean and the OLS estimates lie on the density plots. Briefly discuss the results and comment on the effect of the prior. [2+1 marks]

ii. Show the 95% credibility interval cutoffs on the density plots. Additionally, compute the 95% frequentist confidence intervals from OLS and overlay these on the same plots. Briefly discuss the difference between credibility intervals and confidence intervals. Discuss and compare the computed intervals. [2+2 marks]

iii. Using the credibility intervals, briefly discuss which variables appear to be important predictors of fuel efficiency. Do your findings align with what you would expect? [2 marks]

(d) Consider using the *informative* prior $\tilde{\boldsymbol{\beta}} = (23, -3, -5, -5, 0)^T$ and a prior variance of $\mathbf{M} = 10 \cdot \mathbf{I}_{k+1}$. This reflects a belief that heavier and more powerful cars are less fuel-efficient. Compare the resulting posterior densities to those obtained under the vague prior. Discuss how the choice of prior affects the posterior. Furthermore, discuss the effect of sample size on the posterior. What happens if we only consider a subset of the available data? [3 marks]

Q2) A Bayesian Search [14 marks]

The use of Bayesian methods became extremely popular and instrumental in operations research during the Second World War. In particular, the method of searching for German Submarines grew to rely almost solely on the field of Bayesian Search Theory. We will briefly explore this topic through a question based on a true story.

One evening, there were two fisherman out at sea. They left the boat on autopilot and whilst one was sleeping, the other fisherman fell overboard. When the fisherman still on the boat awoke to find his peer missing, he contacted the coast guard to initiate a search and rescue operation. We will employ a Bayesian search method to find the lost fisherman.

We begin by creating a search grid of the possible areas which the fisherman could be lost. We will use a simple 20×20 search grid where we can search a single cell at a time. Let us define the following:

$$
\begin{aligned}
i &= \text{indices of the grid cells} \\
Y_i &= \text{Binary variable to determine the true state of whether the fisherman is in the } i^{th} \text{ cell,} \\
&\quad \text{where 1 represents that the fisherman is indeed in that cell.} \\
\theta_i &= \text{probability of occurrence} = \mathbb{P}(Y_i = 1)
\end{aligned}
$$

The main problem arises from the difficulty of finding a fisherman floating in the ocean. In particular, there is a *probability of detection* which represents the probability that we find the fisherman given that he is in the cell we are searching. Hence, let

$$
\begin{aligned}
Z_i &= \text{search result of the } i^{th} \text{ cell, 1 meaning detection} \\
p_i &= \text{probability of detection} = \mathbb{P}(Z_i = 1 | Y_i = 1)
\end{aligned}
$$

This means that the two probabilities θ_i and p_i are prior values obtained from experts before the search and rescue begins. We will use Bayes theorem to update these probabilities after each cell search that we conduct.

Let us assume that there is independence between the grid cells and that both the occurrence and detection of the

fisherman in each cell follows a Bernoulli distribution such that:

$$Z_i|Y_i \sim Ber(Y_i p_i) = (Y_i p_i)^z (1 - Y_i p_i)^{1-z},$$
$$Y_i \sim Ber(\theta_i) = (\theta_i)^y (1 - \theta_i)^{1-y}$$

Hence, we are obviously interested in the case where the ith cell contains the fisherman but we fail to detect him. The posterior probability for this case is:

$$\pi(Y_i = 1|Z_i = 0) = \frac{(1 - p_i)\theta_i}{1 - p_i\theta_i} \tag{1}$$

We will use this posterior distribution to update the occurrence probability, θ_i, once we search cell i and have no detection such that:

$$\theta_{i,new} = \frac{(1 - p_i)\theta_{i,old}}{1 - p_i\theta_{i,old}} \tag{2}$$

Similarly, the probability of occurrence in any other cell j, $j \neq i$, has the updating equation:

$$\theta_{j,new} = \frac{\theta_{j,old}}{1 - p_i\theta_{i,old}} \tag{3}$$

What have we achieved? We now have a representation for our posterior distribution to update the probability of the fisherman's location. We can now easily carry out a search and rescue process as such:

1. Gain a prior distribution for the fisherman's location from experts
2. Gain a detection distribution given the location
3. Merge these two distributions to find the distribution of detecting the fisherman in a given cell
4. Perform a search of the cell with the highest probability of a successful detection
5. If the fisherman is not detected, then update the probability distribution via Bayes' Theorem
6. Continue until the fisherman is found

We will simulate this process. Use the provided R code template to carry out this process. The template includes the initial setup of the problem, as well as the prior and detection probability distributions obtained from the expert.

Questions

(a) Clearly derive equations (2) and (4) from the given Bernoulli distributions. [2 marks]

(b) Using the template, simulate the search and rescue process. When you search a cell, simulate the search process by drawing from a Bernoulli trial. The clock is ticking. Record how long it takes to find him (assuming one cell search per hour).

You should produce the following:

(i) Show heatmaps of the occurrence probability θ_i for at least **four stages**. Briefly comment on how the probability mass evolves. [4 marks]

(ii) Track and plot the posterior probability of occurrence for the cell in which the fisherman is truly located, across all search steps. Discuss the pattern you observe. [4 marks]

(c) Now compare the Bayesian search strategy against a **naive strategy** where cells are searched in a random order.

(i) Run both strategies over **100 simulations**. Record the number of steps required to find the fisherman in each simulation. [2 marks]

(ii) Compare the two approaches and provide rationale for better search times. [2 marks]

Q3) A Twist on Linear Regression [15 marks]

Preamble

We are often faced with the problem of outliers. In many cases, statisticians suggest removing outliers from a dataset; however, this is not the only option. A Bayesian approach allows us to explicitly model observations that may have higher variance, accommodating outliers without discarding data.

To see why this matters, we will first contaminate the Auto MPG data from Q1. Use the code below to introduce outliers, then observe what happens to the standard model.

```
# Use the same Auto MPG data from Q1
# Y, X, X.c, n, k should already be in your environment from Q1.
# If not, re-run the Q1 preamble code.

# Contaminate the data with outliers
set.seed(12)
n1 <- 350                      # Number of 'standard' observations
n2 <- n - n1                   # Number of outliers
outlier.idx <- (n1 + 1):n      # Index set I2

# Inflate the response for the outlier observations
Y.contaminated <- Y
Y.contaminated[outlier.idx] <- Y[outlier.idx] +
                                 rnorm(n2, mean = 0, sd = 15)
```

Model

Consider the linear regression model where the data is generated as

$$Y_i = \begin{cases} \boldsymbol{x}_i^T \boldsymbol{\beta} + e_i, & e_i \sim \mathcal{N}(0, \sigma_1^2), i \in \mathcal{I}_1 \\ \boldsymbol{x}_i^T \boldsymbol{\beta} + e_i, & e_i \sim \mathcal{N}(0, \sigma_2^2), i \in \mathcal{I}_2 \end{cases}$$

where

$\mathcal{I}_i$ is an index set

$\sigma_1^2 < \sigma_2^2$.

This implies that observations of the second index set have a higher variance than those of the first, potentially outliers in the data set.

As usual, let $\tau_i = \frac{1}{\sigma_i^2}$. If we assume that $\mathcal{I}_1$ is known, then w.l.o.g., we can let

$$\mathcal{I}_1 = \{1, \ldots, n_1\},$$
$$\mathcal{I}_2 = \{n_1 + 1, \ldots, n\}.$$

where the first n_1 is the number of *standard* observations, and the rest of the observations are outliers.

(a) Derive the following conditional posterior distributions: $\pi(\beta|\tau_1, \tau_2, X, Y)$, $\pi(\tau_1|\beta, \tau_2, X, Y)$, and $\pi(\tau_2|\tau_1, \beta, X, Y)$. For the derivation of the posteriors, assume the following prior distributions: $\beta \sim \mathcal{N}(0, T_0)$, $\tau_1 \sim \mathcal{G}(a, b)$, and $\tau_2|\tau_1 \sim \mathcal{G}(a, b)I_{\tau_1 > \tau_2}$. [9 Marks]

(b) Write R code to obtain samples from $\pi(\beta, \tau_1, \tau_2|X, Y)$. [3 Marks]

(c) Using at least 50 000 posterior samples (after burn-in):

(i) Produce trace plots for $\boldsymbol{\beta}$, τ_1, and τ_2 to assess convergence. [1 Mark]

(ii) Compare these samples to those in Question 1; comment on the difference. [2 Marks]